



S Naveen cs26e010 <cs26e010@smail.iitm.ac.in>

CS5013 project

5 messages

Omkar Dhawal <omkar@cse.iitm.ac.in>

Fri, Aug 28, 2026 at 1:47 PM

To: "cs26e010@smail.iitm.ac.in" <cs26e010@smail.iitm.ac.in>, CS26E005@smail.iit

Cc: Yuvraj Talukdar CS23D009 <cs23d009@cse.iitm.ac.in>, CS25S011 Mitesh Sharma <cs25s011@cse.iitm.ac.in>, cs25d005@cse.iitm.ac.in

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Please send us a detailed response by replying to this mail using "reply all".

Deadline - 1st Sept

Regards
CS5013 TAs

Omkar Dhawal <omkar@cse.iitm.ac.in>

Fri, Aug 28, 2026 at 1:51 PM

To: "cs26e010@smail.iitm.ac.in" <cs26e010@smail.iitm.ac.in>, "cs26e005@smail.iitm.ac.in" <cs26e005@smail.iitm.ac.in>

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S Naveen cs26e010 <cs26e010@smail.iitm.ac.in>

Mon, Aug 31, 2026 at 10:34 PM

To: Omkar Dhawal <omkar@cse.iitm.ac.in>

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(A) concrete cases where it breaks down, and
(B) how we plan to solve it.

A. Where DOI-based dedup is not straightforward

1. The DOI is simply missing.

A large share of scholarly records carry no DOI at all: most pre-2000 literature, many conference/workshop papers, book chapters, theses, technical reports, patents, datasets, and user-entered citations. DOIs only became widespread after ~2000. So for a big slice of the corpus, DOI can never be the join key — the only signal available is noisy metadata (title, authors, venue, year). Any system that leans on DOI silently fails to dedup precisely the hardest records.

2. One work → many DOIs. The "one DOI = one work" assumption often doesn't hold:

- Preprint vs version of record: the arXiv preprint and the published paper carry different identifiers (e.g. an arXiv DOI vs the publisher DOI) for what is essentially the same paper. Are these duplicates? That's a policy decision, not a lookup.
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- Reprints, translations, errata/corrigenda, new editions, and book-chapter DOI vs whole-book DOI — all distinct DOIs for closely related items.

3. The DOI is present but wrong or inconsistent.

DOIs get mangled: casing (10.1000/XYZ vs 10.1000/xyz — spec says case-insensitive, but data isn't stored that way), prefixes (<https://doi.org/...> vs bare), trailing punctuation copied from a citation, URL-encoding, or an outright wrong DOI pasted from a reference list. So even "DOI present" needs normalization and can't be trusted blindly.

4. When DOI is absent, the metadata is genuinely fuzzy — this is the core of the project:

- Titles: "Attention Is All You Need" vs "Attention is all you need" vs the same with a subtitle; LaTeX/HTML/Unicode differences (\$k\$-means vs k-means, Müller vs Mueller vs Muller, ligatures, math symbols); "(Extended Abstract)" / "Part II" suffixes.
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- Venue: abbreviations and renames (NIPS → NeurIPS → Advances in Neural Information Processing Systems).
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5. Scale. Naive pairwise comparison is $O(n^2)$; on a corpus of millions of records that's billions of comparisons — infeasible without a blocking/indexing strategy.

6. Defining and resolving "duplicate." Pairwise decisions must be made transitively consistent — if $A \approx B$ and $B \approx C$ but $A \not\approx C$, we need a principled way to form clusters and choose one canonical "golden" record. And the preprint/extended-version question above means "duplicate" must be defined before it can be measured.

B. How we plan to solve it

We treat this as an entity-resolution pipeline, not a lookup:

1. Canonicalization. Normalize DOIs (lowercase, strip prefixes/whitespace/punctuation); normalize titles (Unicode NFKC, strip LaTeX/HTML, fold case/diacritics, remove stopwords); parse author names into given/family with initial- and diacritic-folding; map venues to a controlled vocabulary via an abbreviation dictionary.
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In short: DOI is our fast path for the easy records, but the interesting $\sim[X]\%$ of the work is the records where DOI is missing/multiple/dirty — and that's where the blocking + fuzzy-matching + clustering machinery earns its keep.

Happy to walk through any part in more detail or share our current results.

Best regards,

S Naveen (CS26E010)

Gowni Vamshi (CS26E005)

Department of CSE, IIT Madras

On Fri, Aug 28, 2026 at 1:52 PM Omkar Dhawal <omkar@cse.iitm.ac.in> wrote:

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Regards
CS5013 TAs

Omkar Dhawal <omkar@cse.iitm.ac.in>

Tue, Sep 1, 2026 at 5:12 PM

To: S Naveen cs26e010 <cs26e010@smail.iitm.ac.in>

Cc: "cs26e005@smail.iitm.ac.in" <cs26e005@smail.iitm.ac.in>, Yuvraj Talukdar CS23D009 <cs23d009@cse.iitm.ac.in>, CS25S011 Mitesh Sharma <cs25s011@cse.iitm.ac.in>, cs25d005@cse.iitm.ac.in

Hello Naveen

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CS21D202

On Mon, 31 Aug 2026, 22:35 S Naveen cs26e010, <cs26e010@smail.iitm.ac.in> wrote:

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S Naveen **cs26e010** <cs26e010@smail.iitm.ac.in>
To: Omkar Dhawal <omkar@cse.iitm.ac.in>

Thu, Sep 3, 2026 at 10:19 PM

Cc: "cs26e005@smail.iitm.ac.in" <cs26e005@smail.iitm.ac.in>, Yuvraj Talukdar CS23D009 <cs23d009@cse.iitm.ac.in>, CS25S011 Mitesh Sharma <cs25s011@cse.iitm.ac.in>, cs25d005@cse.iitm.ac.in

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